TRACK C: CONSUMER, FMCG & RETAIL

Team: **Who cares** | Author: **Abhay Singh & Team**

ARGUS: Autonomous Financial Discovery Agent

The agent finds where reported company growth separates from real business reality, then proves or disproves it.

Deliverables: GitHub Repository & Code • 2-Minute Video Demonstration • Round 1 Submission (ATMOS 2026)

1. IDEA & APPROACH: WHY TRUE DISCOVERIES BEAT BASIC AI SUMMARIES — CORE INSIGHT

Most business AI tools today are just **summarizers**: you paste a 30-page company speech, and the AI repeats whatever good news the CEO said. But company leaders are paid to sound positive. The real business story is almost always hidden in the gap between what executives *say* and what their audited accounts and store numbers *actually show*.

ARGUS is not a chatbot or a summarizer. It operates like an independent investigative auditor through six simple actions: **Discover** (spot unusual gaps), **Connect** (link executive words with official balance sheets), **Calculate** (run exact math with no guesswork), **Reason** (test whether growth is genuine), **Challenge** (play devil's advocate to find fair explanations), and **Prove** (deliver clear findings with clickable public links).

> **Strict Rules:** Zero stock tips or investment recommendations (no "buy", "sell", "hold", or price targets). 100% public information only—nothing paid or behind a password.

2. TOOLS & STACK: REAL TECHNOLOGY (NOT JUST VAGUE "AI") — ARCHITECTURE

Tool / Component	What Powers It	Why It Was Chosen
Public Filings Harvester	Screener.in Financial Tables & BSE India PDF Announcements	100% free, public records of audited quarterly P&L, balance sheet schedules, and investor call transcripts.
Autonomous Agent Framework	Python 3 Pipeline (Urllib, HTML Parsers, RegEx Engine)	Autonomous end-to-end execution in under 5 seconds with zero human intervention and zero brittle setup.
Deterministic Math Engine	Standard Python Arithmetic Module (No LLM Math)	Guarantees 0% math errors. Averages, growth rates, and turnover days are calculated by exact code, not AI imagination.
Reasoning & Red-Teaming	Gemini API (Structured Analytical Prompts)	Formulates investigative questions and stress-tests findings against fair industry counter-arguments.
Single Output Interface	Interactive HTML5 Dashboard with Inline SVG Charts	Clean, responsive single-screen dashboard with clickable source links and one-click PDF Executive Memo printing.

3. WORKFLOW: FROM COMPANY NAME TO VERIFIED DISCOVERY — 5-STEP PROCESS

1. Company — Enter any ticker (e.g. DABUR, TRENT) → **2. Investigation** — Pick weak spots: Price hikes vs Volume? → **3. Research Plan** — Harvest 12 quarters of P&L, notes & calls → **4. Evidence** — Cross-examine 3 legs: CEO + Math + Rivals → **5. Discovery** — 1-sentence finding + exact disproof rule

How it thinks: The agent does not read aimlessly. It immediately asks: *"Are higher sales coming from selling more goods, or did they just raise prices? Are distributors taking longer to pay their bills? Are new store openings hiding slowing sales at older stores?"*

4. TRACK & GENERALITY: WORKS ON EVERY CONSUMER COMPANY — ZERO HARDCODING

ARGUS contains **zero hard-coded company names** and requires no custom tuning between runs. It automatically identifies what kind of business it is inspecting and applies the correct analytical rules:

A. Factory Brands (FMCG Makers)

E.g., Dabur, Britannia, Bikaji (outside NIFTY 50).
Checks if higher revenue is just from raising prices while actual units sold are flat, or if warehouse stock is piling up because distributors cannot sell the goods.

B. Retail Stores & Jewellery (Non-Volume Disclosers)

E.g., Trent (Westside/Zudio), DMart, V2 Retail (outside NIFTY 50).
Supermarkets and clothing chains do not report volume in kilograms. ARGUS tracks whether opening shiny new stores is creating an illusion of growth while older stores slow down.

EVIDENCE, PROOF & VALIDATION

Team: **Who cares**

Cross-Examination, Self-Falsification & The Working Dashboard

5. TRIANGULATION: HOW CLAIMS ARE FACT-CHECKED (THE 3-LEGGED STOOL) TRUTH VERIFICATION

In a court of law, a suspect's statement is never accepted without independent proof. Similarly, ARGUS never trusts a single document. Every claim requires **three independent legs of proof**:

Leg of Evidence	What We Inspect	How a Contradiction is Caught
Leg 1: Leadership Story	CEO statements in investor presentations and quarterly earnings calls.	The CEO announces: *"Customer demand is booming across all regions."*
Leg 2: Audited Math	Official balance sheet notes and quarterly cash reports filed with regulators.	The official balance sheet shows unpaid customer bills (receivable days) jumped +35%, meaning goods are sitting unpaid on distributor shelves.
Leg 3: Industry Rivals	Reports from 2 or more direct competitors during the same quarter.	Competitors show normal payment cycles, proving this slowdown is specific to the company, not a general market trend.

6. SELF-FALSIFICATION: THE "CHANGE-MY-MIND" RULE NO BIAS

Most human analysts suffer from *confirmation bias*—they fall in love with their theory and ignore contrary facts. ARGUS avoids this through two built-in sanity checks:

1. The Fair Alternative (Playing Devil's Advocate)

The agent actively looks for honest business explanations. For example: *"Did inventory rise because management bought raw materials early to protect against upcoming price inflation, or did new stores just need normal opening stock?"*

2. The Exact Disproof Condition (Measurable & Time-Bound)

The system sets an exact milestone that would make it admit it was wrong. Example: *"This concern is disproven if the company delivers **two consecutive quarters** of verified volume growth over +4.0% with unpaid bills shrinking."*

7. OUTPUT FORMAT: SINGLE INTERACTIVE DASHBOARD (THE 8 ELEMENTS) LIVE INTERFACE

The system delivers a single output: an **Interactive Financial Research Dashboard** with an instant "Export Executive Memo" button. It contains all **eight required elements**:

- **1. The Discovery:** Exactly 1 quantified sentence showing the hidden gap.
- **2. Why It Matters:** 3 to 5 clear bullets on financial sustainability.
- **3. Key Numbers:** Exact calculations and inline SVG visual trend charts.
- **4. Timeline:** When the shift started and how it developed over quarters.
- **5. Evidence Trail:** Clickable links with document title, page, date & excerpt.
- **6. Counterargument:** The strongest alternative business explanation.
- **7. Falsification Condition:** Specific numbers that would disprove the finding.
- **8. Evidence Strength:** Strong, Moderate, or Weak with full explanation.

ARGUS DISCOVERY DASHBOARD • Track C: Consumer, FMCG & Retail — **Evidence Rating: [STRONG] • [Print PDF Memo]**

1. THE DISCOVERY: [Company] reports headline revenue expansion (+16.9% YoY), yet underlying growth is driven by new store openings while warehouse stock grew +68.2% (₹2,289 Cr), masking maturing sales at older locations.

2. WHY IT MATTERS (3-5 Bullets)
- New stores mask older store slowdown
- Warehouse stock locks up ₹2,289 Cr cash
- Lease liabilities elevate markdown risk

3. NUMBERS & CHARTS + 4. TIMELINE
[Growth: +16.9%] [Stock Days: 51.2 Days] [DSO: 0.9d]
[Inline SVG Bar Chart: Q1 ₹4,818Cr • Q4 ₹5,755Cr]
Timeline: Jun 2023 baseline → Mar 2026 expansion

5. EVIDENCE TRAIL (Clickable Documents)
[Audited Results]: Screener.in Consolidated P&L
[Earnings Call]: BSE India Filing Transcript PDF
[Competitors]: Direct Peer Financial Disclosures

6. COUNTERARGUMENT & 7. FALSIFICATION
Counter: New stores take 18-24 months to mature.
Disproof Rule: Two consecutive quarters of sales growth >8.0% with stock turnover under 50 days.

8. BUILD PLAN & MILESTONES TO 25 OCT (HONEST CLAIMS VS REALITY) ROADMAP

Date / Milestone	Status: Built vs. Planned	What It Delivers for the Finals
Oct 18 (Today) **Working MVP**	**BUILT & TESTED TODAY** Fully autonomous execution.	Pulls live public filings, runs deterministic math, detects retail/FMCG differences, tests counterarguments, and builds the 8-element dashboard.
Oct 20–21 **Concall Deep Dive**	**PLANNED**	Extracts exact spoken sentences and speaker names from full-length concall PDFs using fast local text matching.
Oct 22–23 **Peer Matrix**	**PLANNED**	Broadens automated cross-checks from 3 peers to 10+ peers across non-NIFTY mid-caps and regional competitors.
Oct 24–25 **Finals Ready**	**PLANNED**	Stress-tested on 50+ Indian consumer tickers to deliver a flawless live 10-minute presentation on whichever company the judges pick.